MILITARY SPECIFICATION

DELAY ASSEMBLY
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification contains requirements, quality assurance provisions and the preparation for delivery criteria for delay assemblies used on M753 8 Inch Projectiles (see 6.1).

2.0 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-V-50433	Vinyl Alcohol - Acetate Resin Solution (VAAR)
MIL-I-6866	Inspection, Penetrant Method of

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-453	Inspection, Radiographic
MIL-STD-1234	Pyrotechnics: Sampling Inspection and Testing

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS (see 6.13)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9287957	Plug, Igniter
9287958	Delay, Washer
9287972	Housing, Delay
9325983	Projectile, PXR6264, Inert with Live Delay
9326185	Delay Assembly, 8 Inch, M753
9347202	Packing and Marking for Delay Assembly for M753 Projectile

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Assembly. The delay assembly shall comply with all requirements of drawing (dwg.) 9326185, with the requirements of all applicable subassembly dwgs., with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the dwg. and herein (see 6.7).

3.2 Materials and components. The contractor shall provide objective evidence (see 6.4) that materials and components (metal and plastic parts) comply with the requirements herein.

3.2.1 Components. Components (metal and plastic parts) shall comply with applicable M650 Projectile dwg. and specification requirements. In addition, one hundred percent dye penetrant inspection of delay housings for cracks or other metal defects and one hundred

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

percent inspection of selected dimensional characteristics of parts shall be performed as specified herein (see 4.4.2.8).

3.2.2 Pyrotechnic materials. Basic pyrotechnic materials (tungsten, barium chromate etc.) shall comply with the requirements of dwg. 9326185.

3.3 Pyrotechnic characteristics.

3.3.1 Composition. The flash, delay and igniter composition shall comply with applicable dwg. and specification requirements. Final acceptance of the delay composition shall be contingent on successfully meeting the flight test delay time requirements of 3.7.1. Unless otherwise specified or approved by the acquiring agency, each lot of finished delay assemblies shall be assembled from a single approved delay composition lot.

3.3.2 Consolidation load. The consolidation load and dwell time of consolidation process shall be as specified on dwg. 9326185 when tested in accordnce with 4.4.3.2.

3.3.3 Particle size. The granulation of the delay composition prior to loading or pelletization shall be as specified on dwg. 9326185 when tested in accordance with 4.4.3.3.

3.3.4 Total volatiles. The total volatiles of the flash, delay and igniter composition shall not exceed the value specified on dwg. 9326185 when tested in accordance with 4.5.1.

3.3.5 Solids content of VAAR. The solids content of vinyl alcohol-acetate resin (VAAR) shall be as specified on dwg. 9326185 when tested in accordance with 4.5.2.

3.4 Crimp strength. The washer crimp shall be capable of withstanding the minimum load specified on dwg. 9326185 when tested in accordance with 4.4.3.6.

3.5 Pyrotechnic porosity test. The pressure recorded at the exit end of the delay assembly, 80 seconds after application of the nitrogen gas, shall not be greater than 17 PSIG above the zero reference. The pressure recorded at the exit end of the delay assembly 110 seconds after application of the nitrogen gas shall be no less than 2 PSIG and no greater than 27 PSIG above the zero reference. Testing shall be performed in accordance with 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

3.6 Radiographic inspection. The delay column shall not be missing or short in length and shall be free of voids. The igniter and/or flash composition shall not be missing. The boron potassium nitrate charge shall not be missing or contain voids greater than 1/16 square inch. Testing shall be performed in accordance with 4.5.4.

3.7 Flight tests.

3.7.1 Delay composition acceptance.

3.7.1.1 Plus 70°F. After conditioning at 70° ± 5°F for a minimum period of 24 hours, the delays shall be flight tested at a proving ground as specified in 4.5.5.1. The average delay time shall be 7.8 seconds plus or minus 0.5 seconds, and the standard deviation of the delay times shall not exceed 0.5 seconds.

3.7.1.2 Plus 145°F. After conditioning at 145° ±5°F for a minimum period of 24 hours, the delays shall be flight tested at a proving ground as specified in 4.5.5.1. The average delay time shall be 6.8 seconds plus or minus .75 seconds, and the standard deviation of the delay times shall not exceed .50 seconds.

3.7.2 Delay assembly acceptance.

3.7.2.1 Plus 70°F. After conditioning at 70° ±5°F for a minimum period of 24 hours, the delay assembly shall be flight tested at a proving ground as specified in 4.5.5.2.1. The average delay time shall be 7.8 seconds plus or minus 0.5 seconds, and the standard deviation of delay times shall not exceed 0.5 seconds.

3.7.2.2 Plus 145°F. After conditioning at 145° ±5°F for a minimum period of 24 hours, the delay assembly shall be flight tested at a proving ground as specified in 4.5.5.2.2. The average delay time shall be 6.8 seconds plus or minus .75 seconds and the standard deviation of delay times shall not exceed .50 seconds.

3.7.2.3 Minus 50°F. After conditioning at minus 50° ±5°F for a minimum period of 24 hours, the delay assembly shall be flight tested at a proving ground as specified in 4.5.5.2.3. The average delay time shall be 9.3 seconds plus or minus .75 seconds and the standard deviation of delay times shall not exceed .80 seconds.

3.8 Workmanship. The delay assembly shall be free of nicks, dents, burrs, or any other obvious physical defects. Parts shall be clean and free of dirt, grease, chips, corrosion or any other foreign matter. Cleaning methods shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. Preparation of metal surfaces shall be as specified on the applicable dwgs. and in the applicable specifications. Where cleaning and preparation are required by assembly details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract or purchase order, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. This specification contains technical provisions for quality conformance inspection only. First article inspection shall not apply.

4.3 First article inspection. Not applicable.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of delay assemblies shall contain:

a. Inert parts from one interfix lot number from one manufacturer.

b. Flash composition from not more than one inspection lot.

c. Delay composition from not more than one inspection lot.

d. Igniter composition from not more than one inspection lot.

e. Boron potassium nitrate pellets from not more than one interfix lot.

4.4.1.1 Delay composition lot. A delay composition lot is defined as that quantity of delay material from which representative samples have been selected, loaded, and forwarded to a proving ground for delay composition flight test acceptance. A delay composition lot may be formed by any of the following methods but shall not exceed 75 pounds.

a. Blending two or more batches of delay powder or pellets together to form a delay composition lot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

b. Designating a batch of delay powder or delay pellets as delay composition lot.

Once samples are selected from the batch or delay composition lot, the batch or lot shall not be subsequently blended.

4.4.2 Examination. See MIL-A-48078 and the Quality Conformance Inspection tables herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Components, Basic Materials, Composition and Pyrotechnic Composition Acceptance	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Objective evidence for 100 percent inspection of components (see 4.4.2.8)		-	3.2	Visual
102.	Objective evidence for basic pyrotechnic materials		-	3.2	Visual
103.	Pyrotechnic characteristics of flash, igniter and delay composition	4.4. 3.1	-	3.3.1 thru 3.3.4	4.4.2.3 thru 4.4.2.7 (see note 1)
104.	Solids content of VAAR	4.4. 3.5	-	3.3.5	4.5.2
105.	Pyrotechnic composition acceptance flight tests at 70°F and 145°F (see note 2)	4.4. 3.7.1	-	3.7.1	4.5.5.1
	NOTES: 1.) Flight tests for pyrotechnic composition acceptance shall be conducted in accordance with CD 105 herein in lieu of CD's 3 thru 5 of 4.4.2.7. 2.) These tests will be performed with delay assemblies assembled on modified M106 8 Inch projectiles furnished by the acquiring agency.				

NOTES:

Source: https://assist.dla.mil - Downloaded: 2022-10-28T03:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Delay Assembly, Prior to Loading Delay Composition	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Total volatiles of flash composition	4.4. 3.4	-	3.3.4	4.5.1
102.	Consolidation load and dwell time of flash composition	4.4 3.2	-	3.3.2	Gage
103.	Flash composition on sidewall		100%	3.1	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Delay Assembly Prior to Loading Igniter Composition	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY
			M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Total volatiles of delay composition	4.4 3.4	-	3.3.4	4.5.1
102.	Consolidation load and dwell time of each increment of delay composition	4.4 3.2	-	3.3.2	Gage
103.	Distance from bottom of large cavity in housing to delay composition (2 places)		100%	3.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Delay Assembly After Loading Igniter Composition	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:	None defined				
Major:					
101.	Total volatiles of igniter composition	4.4 3.4		3.3.4	4.5.1
102.	Consolidation load and dwell time of igniter composition	4.4 3.2		3.3.2	Gage

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

Source: https://assist.dla.mil -- Downloaded: 2020-10-28T06:06Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Delay Assembly After Crimping Washer	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Washer missing (2 places)		100%	3.1	Visual
2.	Crimp missing or incomplete (2 places)		100%	3.1	Visual
3.	Flash or igniter composition missing (2 places)		100%	3.1	Visual
Major:					
101.	Depth to washer, min (2 places)		100%	3.1	Gage
102.	Delay washer crimp strength	4.4.3.6		3.4	Test/4.4.3.6
103.	Porosity test		100%	3.5	4.5.3
104.	Evidence of poor workmanship		100%	3.8	Visual

NOTES:

Source: https://assist.dla.mil - Downloaded: 2026-10-23T02:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Delay Assembly, Prior to Assembly of Igniter Plug	1 OF 1	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Boron-potassium nitrate pellets missing		100%	3.1	Visual
Major:					
101.	Weight of boron-potassium nitrate pellets		100%	3.1	Balance
102.	Distance from end of delay housing tube to ignition charge, min		100%	3.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2022-10-27T08:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Delay Assembly	1 OF 2	9326185
			NEXT HIGHER ASSEMBLY
			M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical:					
1.	Igniter plug missing or improperly positioned		100%	3.1	Visual
2.	Crimp of plug to delay housing missing or body cracked by crimping operation		100%	3.1	Visual
3.*	Flight test at 70 F	4.4. 3.7. 2.1		3.7. 2.1	4.5.5.2.1
4.*	Flight test at 145 F	4.4. 3.7. 2.2		3.7. 2.1	4.5.5.2.2
5.*	Flight test at -50 F	4.4. 3.7. 2.3		3.7. 2.3	4.5.5.2.3
Major:					
101.	Outside diameter of tube, max.		100%	3.1	Gage
102.	Boron potassium pellets missing		100%	3.6	X-Ray/4.5.4
103.	Flash composition missing (2 places)		100%	3.6	X-Ray/4.5.4
104.	Cracks or voids in delay column, including missing increments (2 places)		100%	3.6	X-Ray/4.5.4
105.	Igniter composition missing (2 places)		100%	3.6	X-Ray/4.5.4

NOTES:

*These tests shall be performed with delay assemblies assembled on modified M106 8 Inch Projectiles furnished by the acquiring agency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7 (cont)	Delay Assembly	2 OF 2	9326185
			NEXT HIGHER ASSEMBLY: M753 Proj.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
106.	Delay washer missing (2 places)		100%	3.6	Visual/X-Ray/4.5.4
107.	Evidence of poor workmanship		100%	3.8	Visual
108.	Inside diameter at bottom of crimps		100%	3.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.8	One Hundred Percent Component Inspection		See Below
			NEXT HIGHER ASSEMBLY
			9326185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	*Housing, Delay (dwg. 9287972).				
Major:					
101.	Inside diameter of holes (2)		100%	3.2.1	Gage
102.	Depth of holes (2)		100%	3.2.1	Gage
103.	Pitch diameter of thread		100%	3.2.1	Gage
104.	Minor diameter of thread		100%	3.2.1	Gage
105.	Surface finish on largest inside diameter		100%	3.2.1	Gage
106.	Dye penetrant inspection for cracks and metal defects		100%	3.2.1	MIL-I-6866
	*Delay Washer (dwg. 9287958).				
Major:					
101.	Inside diameter		100%	3.2.1	Gage
102.	Outside diameter		100%	3.2.1	Gage
	*Plug, Igniters (dwg. 9287957).				
Major:					
101.	Outside diameter		100%	3.2.1	Gage

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2022-10-27T02:22Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	DRAWING NUMBER
4.4.2.8 (cont)	One Hundred Percent Component Inspectiion		2 OF 2	See Below
				NEXT HIGHER ASSEMBLY: 9326185

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	NOTES: *These are M650 Projectile parts. In addition to the above 100% inspection on select characteristics, objective evidence (see 6.4) shall be required verifying conformance of parts to applicable M650 dwgs. and specifications.				

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:29Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Packaging	1 OF 1	See 5.0
			NEXT HIGHER ASSEMBLY: N.A.

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major:					
101.	Packaging and packing not in accordance with applicable requirements.		-	5.0	Visual
102.	Parts not clean and free of dirt and other foreign material prior to packaging.		100%	5.0	Visual
103.	Parts improperly packaged in barrier bags.		100%	5.0	Visual
104.	Spacers, filler and cushioning do not provide firm pack.		100%	5.0	Visual/Manual
105.	Exterior pack improper or not securely sealed.		100%	5.0	Visual/Manual
106.	Marking on individual packages or external pack improper, illegible or missing.		100%	5.0	Visual
107.	Evidence of poor workmanship.		100%	-	Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2020-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. Testing is described in the Quality Conformance Inspection tables and herein.

4.4.3.1 Composition. The composition by weight of each lot of flash, delay and igniter composition shall be recorded and witnessed by an authorized quality assurance representative (see 6.8). For information purposes, each lot of flash, delay and igniter composition shall be analyzed by an approved test method and the results recorded.

4.4.3.2 Consolidation load. The contractor shall provide adequate controls to assure the compliance of each lot with the load and dwell time requirements specified on the dwg. For verification, each consolidation load machine shall be calibrated or checked twice daily (at the midpoint and end of each days shift) for consolidation load and dwell time, using a load cell and stop watch or approved calibrated gages. Load consolidation shall not commence on hydraulic presses until oil temperature has stabilized. If at the time of calibration or check, any consolidation load machine fails to meet the load or dwell time requirements, all delay assemblies loaded on that machine since the previous calibration or check shall be rejected.

4.4.3.3 Particle size. Screening of each lot of delay composition for particle size shall be witnessed by an authorized quality assurance representative (see 6.8).

4.4.3.4 Total volatiles. The contractor shall provide adequate controls to assure the compliance of each lot with the specified total volatiles requirement. For verification a sample of delay, igniter and flash composition from each 8 hours production and from each new container shall be selected to perform a total volatiles analysis. If any sample fails to meet the requirements of 3.3.4, the 8 hours production represented by the sample or the new container of material shall be rejected. Testing shall be conducted in accordance with 4.5.1.

4.4.3.5 Vinyl alcohol acetate resin (VAAR) solids determination. The concentration of solids in the VAAR solution may change during storage due to evaporation of solvent. VAAR solids content in the composition can only be obtained if the solution to solids weight ratio of the VAAR is known. Prior to blending the composition, a sample of VAAR solution shall be tested to determine this ratio. The test shall be conducted as specified in 4.5.2.

4.4.3.6 Crimp strength. At the beginning of production an empty delay housing shall be crimped on each crimping device using the same technique as in regular production except that the housing shall not be loaded and delay washers shall not be utilized. After crimping, the delay housing shall be sectioned perpendicular to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

longitudinal axis and the load applied axially to the crimp from the sectioned end using approved equipment and procedures. If the crimp fails to meet the applicable load requirements, the crimping device shall be adjusted until the crimp meets the requirements of 3.4. The crimp strength test shall then be repeated after every 200 delay assemblies to assure that the proper crimp strength is maintained throughout production. If any test delay fails to meet the required crimp strength requirement, all delay assemblies crimped on the applicable crimping device since the previous crimp strength test shall be rejected.

4.4.3.7 Flight tests.

4.4.3.7.1 Delay composition acceptance.

4.4.3.7.1.1 Plus 70°F. Ten rounds shall be used for acceptance testing of delay composition at 70°F. A minimum of 8 delay times shall be required to determine delay time acceptance. If the burning time requirement of 3.7.1.1 is not met or if the delay assembly fails to function or functions in bore, the delay composition lot shall be rejected. Testing shall be conducted in accordance with 4.5.5.1.

4.4.3.7.1.2 Plus 140°F. Ten rounds shall be used for acceptance testing of delay composition at 145°F. A minimum of 8 delay times shall be required to determine delay time acceptance. If the burning time requirement of 3.7.1.2 is not met or if the delay assembly fails to function or functions in bore, the delay composition lot shall be rejected. Testing shall be conducted in accordance with 4.5.5.1.

4.4.3.7.2 Delay assembly lot acceptance.

4.4.3.7.2.1 Plus 70°F. A sample of ten loaded delay assemblies shall be selected at random and forwarded to a Government Proving Ground for flight tests. A minimum of 8 delay times shall be required to determine delay time acceptance. If the sample fails to meet the burning time requirement of 3.7.2.1, or if any delay assembly fails to function or functions in bore, the lot shall be rejected. Testing shall be conducted in accordance with 4.5.5.2.1.

4.4.3.7.2.2 Plus 145°F. A sample of fifteen loaded delay assemblies shall be selected at random and forwarded to a Government Proving Ground for flight tests. A minimum of 12 delay times shall be required to determine delay time acceptance. If the sample fails to meet the burning time requirement of 3.7.2.2 or if any delay assembly fails to function or functions in bore, the lot shall be rejected. Testing shall be conducted in accordance with 4.5.5.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.4.3.7.2.3 Minus 50°F. A sample of fifteen loaded delay assemblies shall be selected at random and forwarded to a Government Proving Ground for flight tests. A minimum of 12 delay times shall be required to determine delay time acceptance. If the sample fails to meet the burning time requirement of 3.7.2.3, or if any delay assembly fails to function or functions in bore, the lot shall be rejected. Testing shall be conducted in accordance with 4.5.5.2.3.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods and procedures.

4.5.1 Total volatiles. Total volatiles shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.5.2 Solids content of VAAR. The solids content of VAAR shall be tested in accordance with MIL-V-50433.

4.5.3 Pyrotechnic porosity test. Each delay assembly shall be conditioned at 145° ± 5°F for a min. of 16 hours and then assembled on an approved porosity tester. Nitrogen gas under a pressure of 1000 PSIG plus or minus 50 PSIG shall be applied to the delay column at the threaded end of the housing. Pressure at the exit end of the delay column shall be recorded at 0.5 seconds, 80 seconds and 110 seconds. The pressure reading at 0.5 seconds shall be recorded and used as the zero reference.

4.5.4 Radiographic inspection. The delay assembly shall be radiographed perpendicular to the longitudinal axis and oriented so that the two distinct delay columns do not overlap on the radiograph. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of inspection. Radiographic equipment, operations and procedures shall be qualified in accordance with MIL-STD-453 (see 6.5). In addition, the sensitivity of the radiographic technique shall be 2 percent and the photographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

4.5.4.1 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive train. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. (Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units.) Accuracy of the densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal operation shall be checked for accuracy monthly by calibrating the densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablets shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.5 Flight tests.

4.5.5.1 Composition acceptance. These tests shall be performed at a Government Proving Ground with delay assemblies assembled on modified M106 8 Inch Projectiles (dwg. 9325983) furnished by the acquiring agency. Testing shall be performed in accordance with an approved Test Program Request (TPR) provided to the Proving Ground Commander and witnessed and evaluated by the cognizant quality assurance element of ARRADCOM (see 6.9). The loaded delay assemblies shall be x-rayed in accordance with 4.5.4 after pressing into the delay adapter and assembled to the modified M106 8 Inch Projectiles. A rocket off cap shall be assembled to the delay assembly and the complete round, including propelling charge, conditioned at $70^{o} \pm 5^{o}F$ or $145^{o} \pm 5^{o}F$, as applicable, for 24 hours minimum. The complete round shall be removed from the conditioning chamber, a dummy fuze assembled, the rocket-off cap removed, and the shell fired from an M110 cannon in less than 10 minutes from removal from the conditioning box. A zone 9, M188 propelling charge shall be utilized in this test. A smear camera shall be mounted near the muzzle to film the shell and determine presence of the delay at the muzzle. The delay time shall be determined by use of radar or an IR Sensor Tracer (Jefferson PG Manufacture Detector Model No. R0A2Y or equivalent). Stopwatch times may be used as a back-up to confirm that the delay

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

functioned but shall not be used in computations for delay acceptance. If the delay expulsion was not observed on radar, infrared or visually and it has been determined the delay was present at the muzzle, the shell shall be recovered to determine if the delay functioned.

4.5.5.2 Delay assembly lot acceptance.

4.5.5.2.1 Plus 70°F. This test shall be conducted as specified in 4.5.5.1 except that the conditioning temperature for the delay assembly, projectile and propellant charge shall be 70° ± 5°F.

4.5.5.2.2 Plus 145°F. This test shall be conducted as specified in 4.5.5.1 except that the conditioning temperature for the delay assembly, projectile and propellant charge shall be 145° ± 5°F.

4.5.5.2.3 Minus 50°F. This test shall be conducted as specified in 4.5.5.1 except that the conditioning temperature for the delay assembly, projectile and propellant charge shall be -50° ±5°F.

5. PACKAGING

5.1 Packaging, packing and marking. Delay assemblies shall be packaged, packed and marked in accordance with dwg. 9347202.

6. NOTES

6.1 Intended use. Delay assemblies covered by this specification are intended for use on Rocket Motor/Bulkhead Assemblies for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Government furnished property (see 6.10).
c. Flight test requirements at Government Proving Ground (see 6.11).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD1423 Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirements	Applicable DID no.
4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

(Copies of DID's required by contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. Data cards, inspection records, test data or other documentary evidence that parts or materials comply with applicable dwg. and specification requirements.

6.5 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, US Army Armament Research and Development Command, Dover, New Jersey 07801, ATTN: DRDAR-QAN-I for review and approval prior to implementation in accordance with requirements of this specification.

6.6 Proving ground test summary.

Test	Sample Size (see 4.4.3.7.1)	Requirements
Delay Composition Acceptance +70°F	10 Delays	Mean & Std Dev of time & performance (see 3.7.1.1)
+145°	10 Delays	Mean & Std Dev of time & performance (see 3.7.1.2)
Delay Composition Acceptance +145°F	15	Mean & Std Dev of time & performance (see 3.7.2.2)
-50°F	15	Mean & Std Dev of time & performance (see 3.7.2.3)
+70°F	10	Mean & Std Dev of time & performance (see 3.7.2.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.

6.7 Handling of delay assemblies. Precautions should be taken during loading, intraplant shipment for in process tests subsequent to loading, and prior to packing to assure delays are not exposed to moisture. Delay assemblies should not be removed from moisture proof packing until such time as the units are to be placed in the conditioning chamber for porosity testing.

6.8 Particle size and composition verification. The contractor shall arrange for verification with Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAM-VQ, Dover, New Jersey 07801.

6.9 Flight test evaluation. Flight test evaluation shall be performed by QEB, NSD, PAD (DRDAR-QAN-Q). A short summary of test results shall be furnished to Evaluation and Metrology Division, PAD (DRDAR-QAM) within 5 days after completion of flight tests.

6.10 Government furnished property. The contracting officer should arrange to have modified M106 8 Inch Projectiles (dwg. 9325983) furnished by the acquiring agency for the flight tests specified in 3.7, 4.4.2.1 and 4.4.2.7.

6.11 Flight tests. The contracting officer should arrange to have the flight tests specified in 4.5.5 performed at a Government Proving Ground.

6.12 Supersession data. This specification includes the requirements of AR-PD-25B dated 31 March 1981.

6.13 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Comand may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 1110-A165)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T02:22Z
Check the source to verify that this is the current version before use.